

Building2Building: A Large Scale Benchmark for Generalizable Real-World Reinforcement Learning

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Keywords: Reinforcement Learning; Generalization; Multi-Task; Transfer Learning; Benchmark; Energy; HVAC

Summary

Reinforcement learning (RL) has achieved strong results in control, yet learned policies remain brittle to changes in dynamics, action spaces, observation spaces, or goals, a critical limitation for real-world deployment. Existing benchmarks offer limited diversity and complexity, making it difficult to rigorously study transfer, multi-task learning, and meta-learning in RL. We introduce Building2Building (*B2B*), a large-scale suite of realistic Heating, Ventilation, and Air Conditioning (HVAC) control environments built on EnergyPlus, a state-of-the-art building simulator. *B2B* is fully compatible with the Gymnasium interface and features a parametric building generator, enabling the systematic generation of diverse building configurations with heterogeneous observation and action spaces. Based on this suite, we define benchmark tasks targeting key open challenges in RL, including goal adaptation, dynamics adaptation, action-space shifts, and cross-domain transfer. By providing a large-scale, diverse, and physically grounded testbed with standardized evaluation protocols, *B2B* enables systematic investigation of generalization and transfer in continuous control. Beyond advancing research on generalization in RL, this new benchmark also carries significant societal implications by enabling improved HVAC control at scale, one of the most energy-intensive systems in buildings.

Contribution(s)

1. A large-scale suite of control environments based on a high-fidelity HVAC simulator for building climate control. The environments exhibit diverse dynamics and heterogeneous observation and action spaces, providing a testbed for studying generalization in RL.

Context: Widely used RL benchmarks are dominated by robotics and video-game domains, typically featuring homogeneous observation and action spaces, which limits the study of generalization across diverse dynamics and control settings.

2. A well-defined benchmarking framework with reproducible tasks designed to evaluate key aspects of generalization in RL including dynamics adaptation, goal adaptation, action-space shift, and cross-domain transfer.

Context: None

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Abstract

Reinforcement learning (RL) has achieved strong results in control, yet learned policies remain brittle to changes in dynamics, action spaces, observation spaces, or goals, a critical limitation for real-world deployment. Existing benchmarks offer limited diversity and complexity, making it difficult to rigorously study transfer, multi-task learning, and meta-learning in RL. We introduce Building2Building (*B2B*), a large-scale suite of realistic Heating, Ventilation, and Air Conditioning (HVAC) control environments built on EnergyPlus, a state-of-the-art building simulator. *B2B* is fully compatible with the Gymnasium interface and features a parametric building generator, enabling the systematic generation of diverse building configurations with heterogeneous observation and action spaces. Based on this suite, we define benchmark tasks targeting key open challenges in RL, including goal adaptation, dynamics adaptation, action-space shifts, and cross-domain transfer. By providing a large-scale, diverse, and physically grounded testbed with standardized evaluation protocols, *B2B* enables systematic investigation of generalization and transfer in continuous control. Beyond advancing research on generalization in RL, this new benchmark also carries significant societal implications by enabling improved HVAC control at scale, one of the most energy-intensive systems in buildings.

1 Introduction

Reinforcement Learning (RL) has achieved strong performance across a wide range of control problems. In most cases, however, these results are obtained by training a policy on a single target task. While effective in controlled settings, this paradigm limits the ability of RL systems to operate in new environments or adapt to related tasks. Adaptation is a key feature for real-world deployments, and enabling RL agents to generalize and adapt across various environments remains a major open challenge.

Studying generalization in RL requires simulated environments that expose agents to many tasks with varying dynamics, observation spaces, action spaces, and objectives. Existing benchmarks such as Meta-World+ [McLean et al. \(2025\)](#) and RL Bench [James et al. \(2020\)](#) provide useful testbeds, but remain largely confined to the robotics domain. Moreover, the variations they introduce are often limited to changes in dynamics parameters (e.g., friction) or task goals such as target positions for navigation or object placement. As a result, these benchmarks offer limited diversity in environment structure and rarely include heterogeneous observation and action spaces. Because of this limitation, many works studying cross-domain adaptation or transfer in RL rely on custom experimental setups. These environments are often designed for a specific method and are difficult to reproduce, which makes comparisons across approaches challenging. Consequently, progress toward generalizable RL agents remains fragmented.

In this work, we propose a new domain for studying generalization in RL: heating, ventilation, and air-conditioning (HVAC) control in buildings. Buildings exhibit large diversity in layouts, materi-

als, climate exposure, HVAC configurations and objectives balancing energy savings and comfort. At the same time, they share common physical principles based on thermodynamics and energy conservation. This combination of diverse dynamics, heterogeneous control interfaces, and shared physical principles makes HVAC control an ideal domain for studying generalization.

We introduce Building2Building (*B2B*), a large-scale suite of RL environments built on EnergyPlus, a state-of-the-art building energy simulator [NREL \(2017\)](#). EnergyPlus provides high-fidelity physical modeling of building thermal behavior. The simulator’s accuracy is highlighted by works such as [Zhang et al. \(2019\)](#), which showed that RL policies trained in calibrated EnergyPlus simulations can be deployed directly in real buildings.

Our environment suite includes a building generator that produces a virtually infinite number of environments derived from base archetypes across several geographic locations in North America. These environments expose low-level HVAC actuators such as fans and dampers, creating a wide variety of challenging control problems with heterogeneous observation and action spaces, and objectives reflecting different comfort-energy trade-offs. These models are automatically converted into Gymnasium-compatible RL environments through a dedicated processing pipeline.

Using this generator, we construct a dataset of over 6000 environments spanning multiple building types, climates, and HVAC system configurations. We further propose well-defined and reproducible benchmark problems that isolate different sources of variation in RL: goal adaptation, dynamics variation, action-space shifts, and cross-domain generalization.

By providing a large and diverse collection of realistic control environments, *B2B* enables systematic and large-scale evaluation of generalization methods in RL. At the same time, it supports research on intelligent HVAC control and helps bridge the gap between fundamental RL research and real-world deployment on energy management systems. HVAC control represents an important societal challenge. Buildings account for roughly one third of global energy consumption, and HVAC systems alone are responsible for nearly half of this usage. Improving HVAC control can therefore reduce energy consumption, lower operational costs, and contribute directly to climate mitigation [IEA \(2022\)](#).

2 Related Work

2.1 Environments and Benchmarks for Generalization in RL

Existing benchmarks for studying generalization in RL are summarized in Table 1, and the different approaches to the generalization problem in RL are further discussed in Appendix A. Most of the benchmarks focus on either video games or robotics tasks such as locomotion and manipulation. As a result, they cover only a limited set of application domains. These problems have been extensively studied over the past decade. While they have led to significant algorithmic progress, it remains unclear whether these improvements generalize beyond the specific domains on which they are evaluated. As highlighted by [\(Liao et al., 2021\)](#), repeatedly evaluating algorithms on the same benchmarks may lead to overfitting to the benchmark itself rather than genuine progress in general RL capabilities. This motivates the need for new environments covering different applications and offering large diversity of tasks.

2.2 HVAC control

Much of the existing literature on RL for HVAC follows a familiar pattern: select a single building, connect it to a simulator, and demonstrate that an agent can reduce energy consumption while maintaining thermal comfort. Surveys show that this type of study has become common practice [Sierla et al. \(2022\)](#); [Al Sayed et al. \(2024\)](#). However, scaling these approaches beyond individual case studies remains difficult. Each new building requires its own digital twin, with sensors, actuators, and dynamics configured for simulation, and policies trained in one building rarely transfer to others.

Table 1: Main benchmarks used to study generalization in reinforcement learning.

Benchmark	Tasks	Domain
MineRL (Guss et al., 2019)	open / task generators	Video game (Minecraft)
Progen (Cobbe et al., 2020)	16 task families	Procedural video games
RLBench (James et al., 2020)	100 tasks	Robotics manipulation
Meta-World (McLean et al., 2025)	50 tasks	Robotics manipulation
DM Control (Tunyasuvunakool et al., 2020)	~20 tasks	Robotics locomotion / control
ManiSkill2 (Gu et al., 2023)	20 task families	Robotics manipulation
Robosuite (Zhu et al., 2020)	9 tasks	Robotics manipulation
MMBench (Hansen et al., 2026)	200 tasks	Games and robotics
Habitat (Szot et al., 2022)	open / configurable	Embodied navigation

Existing environments, summarized in Table 3, Appendix C reflect this limitation. Platforms such as Sinergym Campoy-Nieves et al. (2025), Energym Scharnhorst et al. (2021), BOPTEST Blum et al. (2021), and CityLearn Nweye et al. (2024) have accelerated research but collectively provide only a few dozen building models. Even within this limited set, policies trained in one environment often fail to generalize to others Manjavacas et al. (2024). Transfer-learning approaches, such as fine-tuning across buildings (Kadamala et al., 2024; Xu et al., 2020; Coraci et al., 2024) or continual learning with hypernetworks (Bekal Udayakumar et al., 2025), provide incremental improvements but remain constrained by the lack of large-scale benchmarks. Field studies report similar challenges: a review of more than one hundred deployments finds inconsistent savings and limited reporting on integration costs (Khabbazi et al., 2025).

In contrast, *B2B* enables large-scale studies of RL for HVAC control. This capability enables a new training paradigm for HVAC control. Rather than designing a highly detailed digital twin for each individual building, policies can be trained across a distribution of buildings that includes the target building. Training across many environments reduces the need for highly accurate per-building models and allows policies to reuse experience across environments, reducing the need to train a new controller from scratch for every building.

3 The Building2Building Suite

3.1 Environment Generation

To access a diverse set of environments representative of real buildings, we rely on two complementary sources of building models that produce a large set of heterogeneous environments, summarized in Table 2.

- A dataset generated by a residential building generator (Larochelle Martin et al., 2026). This generator produces single-zone houses representing the statistical characteristics of the Québec residential building stock.
- A parametric building model generator based on the reference commercial buildings from the ASHRAE 90.1 across 16 different locations in North America. The generator follows the methodology used by the Building Technology Assessment Platform (Shirzadi, 2025), and samples the building size, window-to-wall ratio, insulation levels, and air infiltration. To ensure physical consistency, the generator enforces climate-dependent parameter bounds consistent with ASHRAE standards.

3.2 Control Interface

B2B exposes low-level HVAC actuators, and handles the following systems:

Table 2: Base environments included in the benchmark suite.

Building type	HVAC type	Zones	Action dim.	Observation dim.
House	Unitary system	1	2	7
Restaurant	Unitary systems	3	6	9
Warehouse	Unitary systems & baseboard	3	5	9
Retail Store	Unitary systems	4	8	10
Small Office	Unitary systems	5	5–10	11
Medium Office	Central systems	15	15–33	21

117 1. **Unitary HVAC systems.** These systems condition a single thermal zone using a dedicated unit
 118 such as a heat pump or packaged rooftop unit. For these systems, the agent controls the supply
 119 air temperature (SAT) and the supply air flow rate.

120 2. **Central air loop systems for multi-zone conditioning.** These systems use centralized heating
 121 and cooling equipment that distributes conditioned air through an air loop serving multiple zones.
 122 The agent controls the central SAT, the damper opening of each zone, and the zone-level reheat
 123 coils through thermostat setpoints.

124 3. **Baseboard and heat-only systems.** These systems provide heating only and are controlled
 125 through the zone thermostat.

126 At each timestep, the agent has access to the following observation: weather data (outdoor air
 127 temperature ($^{\circ}C$) and humidity (%), calendar information (time of day, day of week, day of
 128 year), indoor zones air temperature ($^{\circ}C$), and HVAC energy consumption per simulation time step
 129 ($Wh/m^2/timestep$). Note that the energy consumption is normalized by the floor area of the build-
 130 ing to be invariant to building size variations

131 3.3 Control Tasks and Reward Functions

132 To study different trade-offs between thermal comfort and energy efficiency, we propose two reward
 133 functions defining a family of control tasks parametrized by energy savings and comfort:

1.
$$r_t = -\frac{1}{Z} \sum_{z=1}^Z \left(\mathbf{1}_{T_z \in \Delta T} (T_z - T_z^{target})^2 + (1 - \mathbf{1}_{T_z \in \Delta T}) |T_z - T_z^{target}| \right) - w_e \mathcal{E}_{HVAC}(t), \quad (1)$$

134 where $\mathbf{1}_{T_z \in \Delta T}$ is the indicator function of T_z lying within the deadband interval around the target
 135 temperature T_z^{target} , Z denotes the number of zones, and w_e controls the penalty on HVAC en-
 136 ergy consumption $\mathcal{E}_{HVAC}$. Note that the target temperature T_z^{target} may vary over time depending
 137 on occupancy schedules.

2.
$$r_t = -\frac{1}{Z} \sum_{z=1}^Z \left((1 - \mathbf{1}_{T_z \in \Delta T}) (1 + w_T) |T_z - T_z^{target}| \right) - w_e \mathcal{E}_{HVAC}(t), \quad (2)$$

138 where w_T is a large weight applied to deadband violations. This formulation approximates a
 139 barrier function that discourages temperature deviations outside the comfort interval while pro-
 140 moting low-energy operation in the deadband.

141 3.4 Structured environment representation

142 A major obstacle in developing transfer and multi-task RL algorithms is the lack of a standard way
 143 to represent heterogeneous environments. While the standard Gymnasium interface (observation

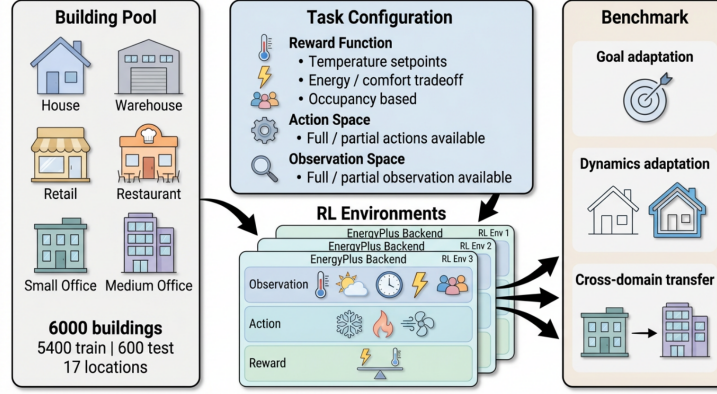


Figure 1: Overview of the benchmark design. A large building pool is created using our building generator. Different benchmark tasks evaluate transfer across goals, dynamics, and domains

space, action space, and step function) is sufficient for single environments, it does not provide a structured representation that can describe multiple environments with different control interfaces. Without such a representation, comparing multi-environment algorithms becomes difficult as each algorithm must implement custom logic to handle each environment separately. In an effort to facilitate cross-domain transfer research, *B2B* introduces a structured representation that provides a common abstraction across heterogeneous environments.

We model the control structure of an environment as a *morphology graph*. A morphology is defined as a graph (V, E) where each vertex $v \in V$ has a type $t(v) \in T$ and where each node type $t \in T$ is associated with a local observation space $S(t)$ and a local action space $A(t)$. The global observation and action spaces of the environment are obtained by concatenating the local spaces of all nodes in a pre-defined order. Formally, we define a morphological universe as $(T, S(t), A(t))$, which is shared and fixed across a collection of environments. A structured environment consists of

1. A standard environment (S, A, step)
2. A morphology graph (V, E)
3. A node type assignment $t : V \rightarrow T$
4. A global-to-local observation mapping : $\text{split} : S \rightarrow \prod_{v \in V} S(t(v))$
5. A local-to-global action mapping : $\text{join} : \prod_{v \in V} A(t(v)) \rightarrow A$

This representation allows policies to operate on the graph structure of the environment and naturally supports architectures such as graph neural networks, heterogeneous graph Transformers [Hao et al. \(2024\)](#), and type-heterogeneous encoder-decoder models [Ma et al. \(2020\)](#). Each environment of *B2B* provides an instance of this structured environment representation. This representation encodes thermal zone adjacency information, groups together actuators within the same HVAC system (e.g., actuators belonging to the same air loop), and establishes the mapping between actuators and the zones they control.

B2B also provides procedures to map between the global and local spaces as described previously. Any algorithm that uses environments through the morphology graph and the `split/join` interface can therefore operate on any *B2B* environment without modification. This enables transfer and multi-task algorithms to be evaluated consistently across buildings with different structures, observation spaces, and control interfaces.

4 Benchmark Problems

To define a reproducible experimental framework, we construct a diverse collection of environments by sampling buildings from the two generators described in Section 3. For each building type, we generate 1000 environments, resulting in a total dataset of 6000 environments, split in a training and a test set. The resulting environments suite spans a wide range of building dynamics, HVAC configurations, and control interfaces. Additional details on the environments generation is given in Appendix B.

Using these environments, we define four benchmark settings that isolate different sources of variation in control systems: goal adaptation, dynamics variation, action-space shifts, and cross-domain generalization. These benchmarks are easily accessible through the API interface of *B2B*.

- **Goal adaptation.** The training and test environments correspond to the same building and control interface, but the reward function changes. We define the following pre-defined rewards:

Task	Reward	w_e	ΔT	Temperature targets
Task 1	Eq. 1	0.01	1.0°C	Constant 21°C
Task 2	Eq. 1	0.10	1.0°C	Constant 21°C
Task 3	Eq. 1	0.01	1.0°C	21°C (occupied), 18°C (unoccupied)
Task 4	Eq. 2	$w_T = 10$	1.0°C	Constant 21°C

- **Dynamics adaptation.** The reward function and action space remain fixed, while the building dynamics vary. Buildings differ in climate zone, size, and envelope but share the same archetype.

Difficulty	Building archetype	Variation	Action dim.
Easy	Single-zone house	Climate, building envelope parameters	2
Medium	Small office	Climate, building envelope parameters	10
Hard	Medium office	Climate, building envelope parameters	33

- **Action-space transfer.** The building dynamics and reward function remain fixed, but the set of controllable actuators changes between training and testing.

System type	Training control	Test control	Action dim.
Unitary system	Air flow rate	Air flow + SAT	5 → 10
Central system	VAV boxes only	VAV boxes + central SAT	30 → 33
Unitary system	Air flow + SAT	Air flow rate	10 → 5
Central system	VAV boxes + central SAT	VAV boxes only	33 → 30

- **Cross-domain generalization.** Agents are trained and tested on different environments:

Difficulty	Train building type	Test building type	Key difference
Easy	Retail store	Small office	Similar HVAC types
Medium	Retail store	Warehouse	Slightly different HVAC types
Hard	Small office	Medium office	Different HVAC types
Hard	n types	m different types	Different domains

5 Baselines

To establish a baseline, we implement supervisory control logic inspired by the ASHRAE Guideline 36 high-performance sequences of operation. Two controllers are implemented: one for unitary systems and one for central air-loop systems. The control structures include zone-level PI loops that regulate airflow and Trim-and-Respond mechanisms that adjust the SAT. Additional

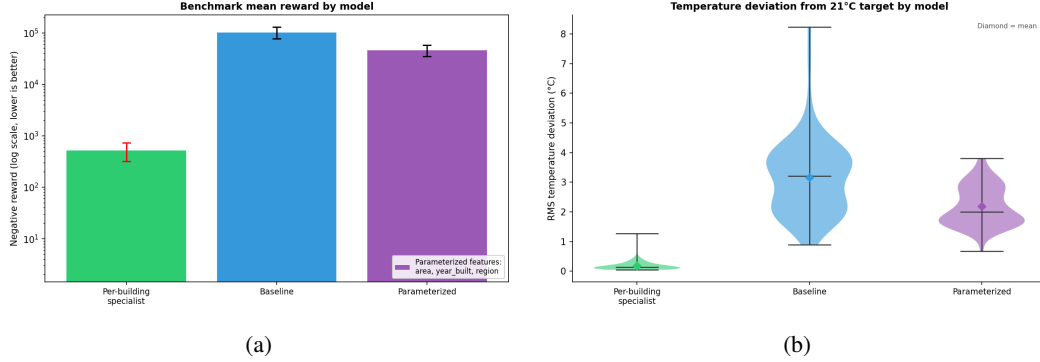


Figure 2: (a) Average reward across 71 test buildings (negative log scale). Lower is better. We show standard error across 9 seeds for baseline and parameterized, and across the 71 test buildings for the per-building specialists. (b) Violin plot of the RMS of temperature deviation from target temperature by model across 71 test buildings. Diamonds represent the overall mean of the distribution.

implementation details are provided in Appendix D, and detailed control results are reported in Appendix E.1. Reactive control strategies are widely used in real buildings, and we therefore use them as the primary baseline. To facilitate evaluation, the benchmark includes a function `compute_normalized_score` that divides the cumulative return over one year of simulation by the return obtained by the reactive controller.

In addition, we train PPO agents on a subset of the test set for each building type. The subset contains 8 buildings, one per climate zone. For single-zone houses, which all belong to the same climate zone, 8 buildings are randomly sampled from the test set. Agents are trained on the four goal adaptation tasks defined in Section 4. Detailed results and normalized scores are reported in Appendix E.2.

6 Generalization Experiments

6.1 Dynamics Adaptation

In order to assess the performance of current methods on dynamics adaptation, we first sample 900 single-zone house buildings for our training set and 71 single-zone buildings for our test set. We then train three different classes of models with PPO: (1) 71 per-building specialists that are trained solely on each building in the test set to establish an upper bound, (2) a centralized agent that resamples new buildings from the training set after every episode, and (3) a centralized agent that resamples new buildings and has its observation space augmented with the building’s floor area, year built, and region. Implementation details are given in Appendix E.3.

Figure 2a shows the average reward across the test buildings for each class of model (with per-building results only on the building that model was trained on). As expected, the per-building specialist has the highest reward, with a substantial gap of $\sim 45,000$ return from the parameterized model and $\sim 105,000$ from the baseline model. The gap between the baseline and parameterized models shows that multi-task learning is essential for solving this cross domain transfer task as there is a large benefit from conditioning on building parameters. The gap between the parameterized and per-building specialist models shows that there is still much room to improve; current multi-task transfer methods are not yet fully capable of achieving the optimal policy when transferred to new test buildings. Figure 2b shows a violin plot of the temperature deviation from the target, showing that the per-building specialists and parameterized models successfully learn to control the buildings. The baseline model performs well for some buildings (likely those that are more similar to ones it encountered in its training set) but performs very poorly for others, demonstrating the necessity of task conditioning.

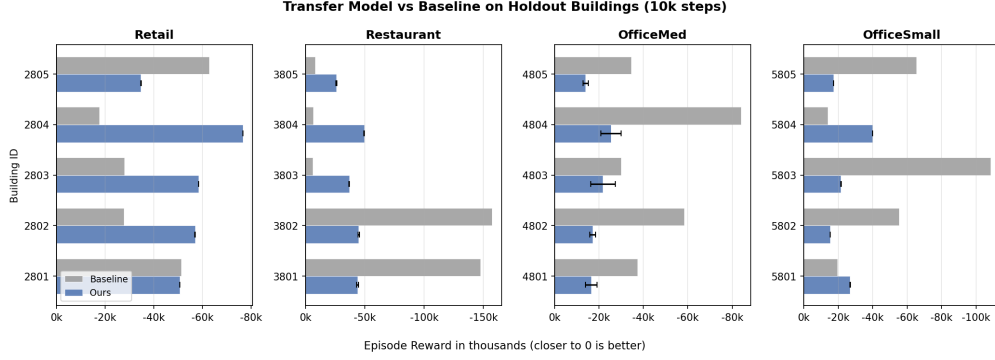


Figure 3: Generalized policy test performance on 20 unseen buildings sampled from the test set.

6.2 Cross domain transfer

To demonstrate how *B2B* can be used to study cross-domain control algorithm, we train a single policy simultaneously on four building types: retail store, fast food restaurant, small office and medium office. These types of buildings have heterogeneous action and observation spaces, making the environment incompatible with the naive use of customary multi-layer perceptron with fixed input and output sizes. We thus implement a variant of the Amorpheus [Kurin et al. \(2021\)](#) architecture that we train on top of the structured environment representation described in Section 3.4.

The policy architecture consists of standard acausal transformer layers together with type-specific linear encoder and decoder blocks. At inference time, the observation is first split into node-level observations, which are encoded into a shared embedding space. The resulting embeddings are processed by the transformer, decoded into node-level actions, and finally concatenated to produce the global actuator command. Algorithm details are given in Appendix E.4.

The model is trained to solve the Task 3. Training data are collected in parallel on four buildings (one of each type). Buildings are resampled every 10 PPO epochs ensuring that the policy is exposed to the full diversity of buildings in the training set. The model is trained on 1M environment steps and tested on new buildings sampled from the test set. As shown in Figure 3 the policy is able to adapt to new domains, and even outperform the baseline reactive controller on some buildings.

7 Conclusion

We introduce *B2B*, a large-scale suite of environments and benchmark problems designed to advance research on generalization in RL. *B2B* provides a diverse set of environments with varying dynamics and heterogeneous observation and action spaces, making it well suited for studying multi-task learning and cross-domain transfer. The benchmark tasks cover several forms of adaptation, including goal adaptation and action-space shifts, and provide a reproducible framework for comparing algorithms.

Grounded in real-world applications, the control tasks focus on HVAC systems in buildings. Improving control in this domain has the potential to reduce energy consumption and contribute significantly to climate change mitigation.

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Supplementary Materials

The following content was not necessarily subject to peer review.

A Prior work on generalization in RL

Generalization in RL has been widely studied. A common approach is Meta RL, where an agent is trained on a distribution of tasks so it can quickly adapt to new ones. Early methods include Model-Agnostic Meta-Learning (MAML) (Shirzadi, 2025), which adapts policies with a few gradient steps, and RL² (Duan et al., 2016), which embeds the learning algorithm inside a recurrent network to share information across tasks via the RNN’s hidden state. Later work introduced adaptation mechanisms such as attention-based meta-learners (Mishra et al., 2018), and probabilistic context inference methods such as PEARL (Rakelly et al., 2019), and Bayes-Adaptive RL (VariBAD) (Zintgraf et al., 2020).

A related line of work studies generalization through a multi-task formulation, where a single model is trained on multiple tasks simultaneously. Such an approach is challenging because gradients from different tasks interfere during learning. Methods such as PCGrad (Yu et al., 2020) mitigate gradient conflicts between tasks during training, while other approaches rely on multiple learner, such as policy distillation (Distral) (Teh et al., 2017) and mixture of orthogonal experts (Hendawy et al., 2024). Large-scale systems such as IMPALA (Espeholt et al., 2018) and BRC (Nauman et al., 2025) demonstrated positive transfer across many tasks, while normalization techniques such as PopArt help address reward-scale imbalance across tasks (Hessel et al., 2018).

Recent works also studies adaptation using sequence modeling and in-context reinforcement learning. In these approaches, models are trained on large datasets of trajectories and adapt to new tasks through conditioning on past experience in context. Seminal works in this direction include Decision Transformer, (Chen et al., 2021), Decision Pretrained Transformer Lee et al. (2023) and Algorithm Distillation (Laskin et al., 2022).

Most multi-task RL methods assume shared observation and action spaces across tasks. However, real-world problems often involve changes in morphology or environments. Methods such as DERL (Gupta et al., 2021), MetaMorph (Gupta et al., 2022), and AnyMorph (Trabucco et al., 2022) study control across agents with different morphologies. Several works study theoretically grounded transfer across domains. For instance successor representations and successor features enable reuse of value structure across related tasks (Barreto et al., 2017; Dayan, 1993). Other approaches provide principled alignment across domains, for example through optimal-transport-based imitation learning (Fickinger et al., 2022).

B Building dataset generation

We generate a climate-consistent building dataset using climate-dependent parameter ranges derived from ASHRAE 90.1-2022 Tables 5.5-1 through 5.5-8. Instead of sampling all seven envelope and geometry parameters independently over uniform global bounds, each building’s location is first mapped to its ASHRAE climate zone, and LHS unit samples are then remapped to zone-specific ranges defined relative to the prescriptive maxima for fenestration and framed walls. Window U-factors are sampled within $[0.8, 1.3] \times$ the ASHRAE maximum for the given zone; SHGC is sampled between 0.15 and $(\text{SHGC}_{\max} + 0.10)$, clamped to $[0.10, 0.80]$; and envelope conductivity and infiltration multipliers are upper-bounded as a function of climate stringency, ensuring tighter constructions in colder zones (ASHRAE 90.1-2022, §5.4.3.1; Tables 5.5-1–5.5-8). Geometry scaling is narrowed to $[0.7, 1.5]$ to avoid extreme HVAC autosizing artifacts, while orientation remains uniformly sampled in $[0^\circ, 360^\circ]$. LHS sampling is preserved within each climate zone by first generating unit hypercube samples and then mapping them through zone-specific bounds. Finally, balanced train/test splits are produced with a dedicated script that performs stratified sampling by location (90/10), ensuring proportional climate representation across splits.

Table 3: Comparison of building control environments

Environment	Buildings
RL Testbed Moriyama et al. (2018)	1
BOPTEST(-Gym) Blum et al. (2021)	~12
Energygym Scharnhorst et al. (2021)	14
Sinergym Campoy-Nieves et al. (2025)	~30
Smart bldg. Control Goldfeder et al. (2025)	11
CityLearn Nweye et al. (2024)	Variable
Building2Building (Ours)	Thousands

499 C Environments details

500 Building2Building converts building simulation models into standardized RL environments. Start-
 501 ing from building descriptions encoded as EnergyPlus input files, *B2B* automatically constructs a
 502 controllable simulation environment compatible with the Gymnasium API ([Towers et al., 2024](#)).
 503 The pipeline parses the building model, identifies thermal zones and HVAC systems, exposes rel-
 504 evant control actuators, and generates standardized observation variables. This process abstracts
 505 away simulator-specific details and produces environments with consistent interfaces for observa-
 506 tions, actions, and rewards. As a result, RL agents can be evaluated across a diverse set of buildings
 507 without requiring any simulator-specific implementations.

508 The resulting Gymnasium environment interacts with the EnergyPlus simulation using callbacks at
 509 every step to fetch observations and pass actions. An EnergyPlus timestep takes about 0.1ms, and
 510 multiple EnergyPlus simulations can be launched in parallel. Timestep durations for the different
 511 building types are given in Table 4.

512 Other HVAC control environments based on high-fidelity open-source models, such as Synergym
 513 [Campoy-Nieves et al. \(2025\)](#), provide only a limited number of buildings. This limitation arises
 514 from two main challenges. First, EnergyPlus simulations require detailed input building descrip-
 515 tions, which are rarely available at large scale. Second, interacting with a complex simulator such
 516 as EnergyPlus in a consistent way is difficult, especially when exposing low-level HVAC actuators
 517 for control. *B2B* addresses these challenges in two ways. First, it provides a generator of Ener-
 518 gyPlus building files that enables the creation of numerous building environments in a principled
 519 way, allowing sensors and actuators to be extracted automatically. Second, it includes a processing
 520 pipeline that automatically converts EnergyPlus models into Python RL environments following the
 521 Gymnasium interface, allowing direct use with existing RL algorithms.

Table 4: EnergyPlus simulation step time per building type, computed over 10000 steps

Building Type	Mean (ms/step)	Std (ms/step)
Office Small	0.170	0.018
Office Medium	0.410	0.096
Retail Standalone	0.153	0.021
Restaurant Fast Food	0.059	0.006
Warehouse	0.209	0.044
Single-Family House	0.021	0.002

522 D Baseline controllers

523 D.1 Reactive Control Logic for Unitary Systems

524 For zones equipped with a unitary system, the action space is the supply fan air mass flow rate and
 525 the supply air temperature setpoint. The control logic follows supervisory principles of ASHRAE
 526 Guideline-36, adapted to a 5 minutes timestep. The zone temperature is regulated using a PI con-
 527 troller that modulates the supply air flow rate. Let T_z denote the zone air temperature and T_{sp} the
 528 active temperature setpoint (cooling or heating). The control error is defined as

$$e(t) = T_z(t) - T_{sp}(t). \quad (3)$$

529 The airflow command is

$$\dot{m}_{sa}(t) = \dot{m}_{\min} + \left[K_p e(t) + K_i \int_0^t e(\tau) d\tau \right] (\dot{m}_{\max} - \dot{m}_{\min}), \quad (4)$$

530 subject to saturation:

$$\dot{m}_{sa} \in [\dot{m}_{\min}, \dot{m}_{\max}]. \quad (5)$$

531 The gains K_p and K_i are tuned for stability under a 15-minute timestep. When the zone temperature
 532 lies within the deadband, airflow is reduced to $\dot{m}_{\min}$.

533 The supply air temperature setpoint is adjusted using a Trim-and-Respond logic based on the zone
 534 demand. For instance, in a cooling regime:

$$T_{sa,sp}^{k+1} = \begin{cases} T_{sa,sp}^k - \Delta_{\text{resp}}, & \text{if } e_c^k > \delta \\ T_{sa,sp}^k + \Delta_{\text{trim}}, & \text{otherwise} \end{cases} \quad (6)$$

535 subject to bounds:

$$T_{sa,sp} \in [T_{sa,\min}, T_{sa,\max}]. \quad (7)$$

536 Here Δ_{resp} and Δ_{trim} represent the respond and trim increments, respectively.

537 D.2 Reactive Control Logic for Air Loops

538 For zones served by a central air loop, the action space consists of the central supply air temperature
 539 setpoint $T_{sa,sp}$ and the per-zone damper opening fraction $\alpha_i \in [0, 1]$. Each zone i is equipped with
 540 a thermostat that modulates the local reheat coil. The zone temperature is regulated through the
 541 damper position using a PI controller. Let $T_{z,i}$ denote the zone air temperature and $T_{sp,i}$ the active
 542 temperature setpoint. The control error is

$$e_i(t) = T_{z,i}(t) - T_{sp,i}(t). \quad (8)$$

543 The commanded damper opening fraction is

$$\alpha_i(t) = \alpha_{\min} + \left[K_p e_i(t) + K_i \int_0^t e_i(\tau) d\tau \right] (\alpha_{\max} - \alpha_{\min}), \quad (9)$$

544 subject to

$$\alpha_i \in [\alpha_{\min}, \alpha_{\max}]. \quad (10)$$

545 When the zone temperature lies within the deadband, the damper position is maintained at $\alpha_{\min}$.
 546 Each zone thermostat activates the reheat coil when the zone temperature falls below the heating
 547 setpoint. The reheat power $Q_{rh,i}$ is modulated proportionally to the heating error

$$Q_{rh,i}(t) = K_{rh} [T_{sp,i}(t) - T_{z,i}(t)]_+, \quad (11)$$

548 where $[x]_+ = \max(0, x)$.
 549 The central supply air temperature setpoint is adjusted using a Trim-and-Respond logic based on
 550 aggregate cooling demand:

$$T_{sa,sp}^{k+1} = \begin{cases} T_{sa,sp}^k - \Delta_{\text{resp}}, & \text{if } D_c^k > \delta \\ T_{sa,sp}^k + \Delta_{\text{trim}}, & \text{otherwise} \end{cases} \quad (12)$$

551 subject to

$$T_{sa,sp} \in [T_{sa,\min}, T_{sa,\max}]. \quad (13)$$

552 D.3 PPO implementation

553 We use Stable-Baseline3 [Raffin et al. \(2021\)](#) for the PPO implementation. Rollouts are collected on
 554 14 environments in parallel. The training is done over 5M environment steps. The hyperparameters
 555 used are summarized in Table 5.

Table 5: Hyperparameters used for PPO training.

Parameter	Value	Parameter	Value
Policy type	MLP	Learning rate	5×10^{-5}
Batch size	336	n_steps	672
Target KL	0.02	n_epochs	5
γ	0.98	λ (GAE)	0.95
Clip range	0.2	Entropy coef.	0.01
Value function coef.	0.5	Max grad norm	0.5
Use SDE	False	SDE sample freq.	96
Policy network architecture			
Actor layers	[256, 256]	Critic layers	[256, 256]
Activation	Tanh	Orthogonal init	True
Initial log std	-1.0		

556 E Additional experiments

557 E.1 Reactive control logic

558 In this section we present the results of the reactive controller on the test buildings. The parameters
 559 of each controller were tuned on one instance of each building type per climate zone using Bayesian
 560 optimization, with Task 1 as the optimization objective. The resulting parameters were then used for
 561 all other buildings of the same type and location.

Table 6 presents the average performance of the reactive controllers. Most of the time, the temperature in each controlled zone remains within the target interval. Failure to maintain temperatures within the controller’s deadband may result from suboptimal control actions, but also from building design and weather conditions that make temperature control impossible. Figure 4 and Figure 5 illustrate the differences in performance across buildings. Figure 7 highlights the impact of the climate zone on control quality. Hotter zones are harder to control, primarily because of HVAC sizing.

Table 6: Percentage of time that zone temperatures remain inside the deadband across test buildings.

Building	Mean	Median	Std
Small Office	92.86	97.55	10.85
Medium Office	83.8	93.17	15.11
Fast-Food Restaurant	82.82	92.93	21.38
Retail Store	88.98	89.23	7.92
Warehouse	90.44	91.42	7.28
Single-Zone House	83.30	85.73	15.69

E.2 PPO control

Table 7 presents the results of the PPO agents on each test building type, average over the locations. The normalized score is computed by deviding by the reactive control baseline. As we can see, the RL agent does not perform as well as the baseline. Figure 6 presents the temperature distribution for two of the four tasks. As we see, the mean temperature is close to the target temperautre, however, a wider distribution causes accumulated penalties that results in lesser return than the baseline.

Building	Task	Mean Temp. In Band (%)	Normalized Return
Small Office	Task 1	47.06	6.04
	Task 2	40.56	4.23
	Task 3	35.45	4.34
	Task 4	46.89	15.77
Medium Office	Task 1	44.85	4.69
	Task 2	46.29	3.27
	Task 3	37.76	
	Task 4	45.95	
Fast-Food Rest.	Task 1	36.44	4.51
	Task 2	20.21	2.72
	Task 3	34.00	3.68
	Task 4	24.51	1.81
Standalone Retail	Task 1	43.73	3.64
	Task 2	36.86	2.46
	Task 3	17.75	4.00
	Task 4	37.89	5.46
Warehouse	Task 1	35.11	3.96
	Task 2	37.75	3.38
	Task 3	58.84	1.00
	Task 4	54.68	3.75

Table 7

575 E.3 Dynamics adaptation

576 All buildings have a two dimensional action space and their observation space is padded to 20
 577 dimensions (some differ due to varying numbers of unconditioned zones. Each model is trained
 578 with 4 million timesteps, with each episode spanning 8928 timesteps (1 month) in the winter period.
 579 (2) and (3) were trained with 16 different buildings running in parallel that were resampled from the
 580 training set with each episode. Policy and value networks for all models were neural networks with
 581 tanh activation and 2 hidden layers with 256 neurons.

582 E.4 cross domain transfer

583 let $\text{Emb} = \mathbb{R}^n$ denote the embedding space for a fixed embedding dimension n . The learnable
 584 components of the policy are (i) type-specific linear encoders $\{e_t : S(t) \rightarrow \text{Emb}\}_{t \in T}$, (ii) a multi-
 585 layer transformer $\text{trans} : \text{Emb}^{|V|} \rightarrow \text{Emb}^{|V|}$, and (iii) type-specific linear decoders $\{d_t : \text{Emb} \rightarrow$
 586 $A(t)\}_{t \in T}$. Policy evaluation proceeds as follows:

```

587 1: function POLICY( $s_{\text{raw}}$ )
588 2:    $s_v \leftarrow \text{split}(s_{\text{raw}})_v \quad \forall v$ 
589 3:    $i_v \leftarrow e_{t(v)}(s_v) \quad \forall v$ 
590 4:    $o \leftarrow \text{trans}(i)$ 
591 5:    $a_v \leftarrow d_{t(v)}(o_v) \quad \forall v$ 
592 6:   return join( $a$ )
593 7: end function

```

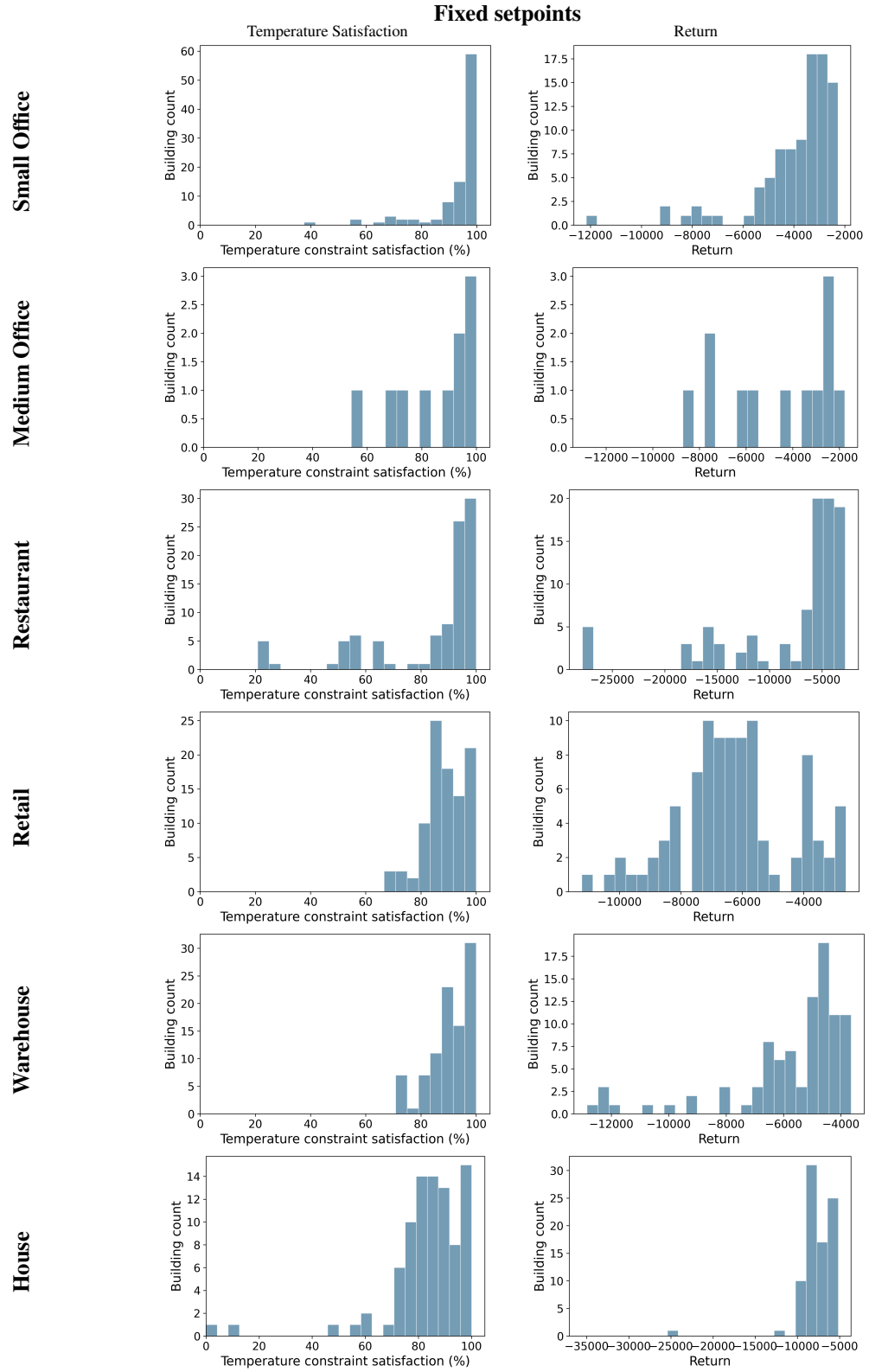


Figure 4: Histograms of temperature satisfaction and returns for fixed setpoints (task 1).

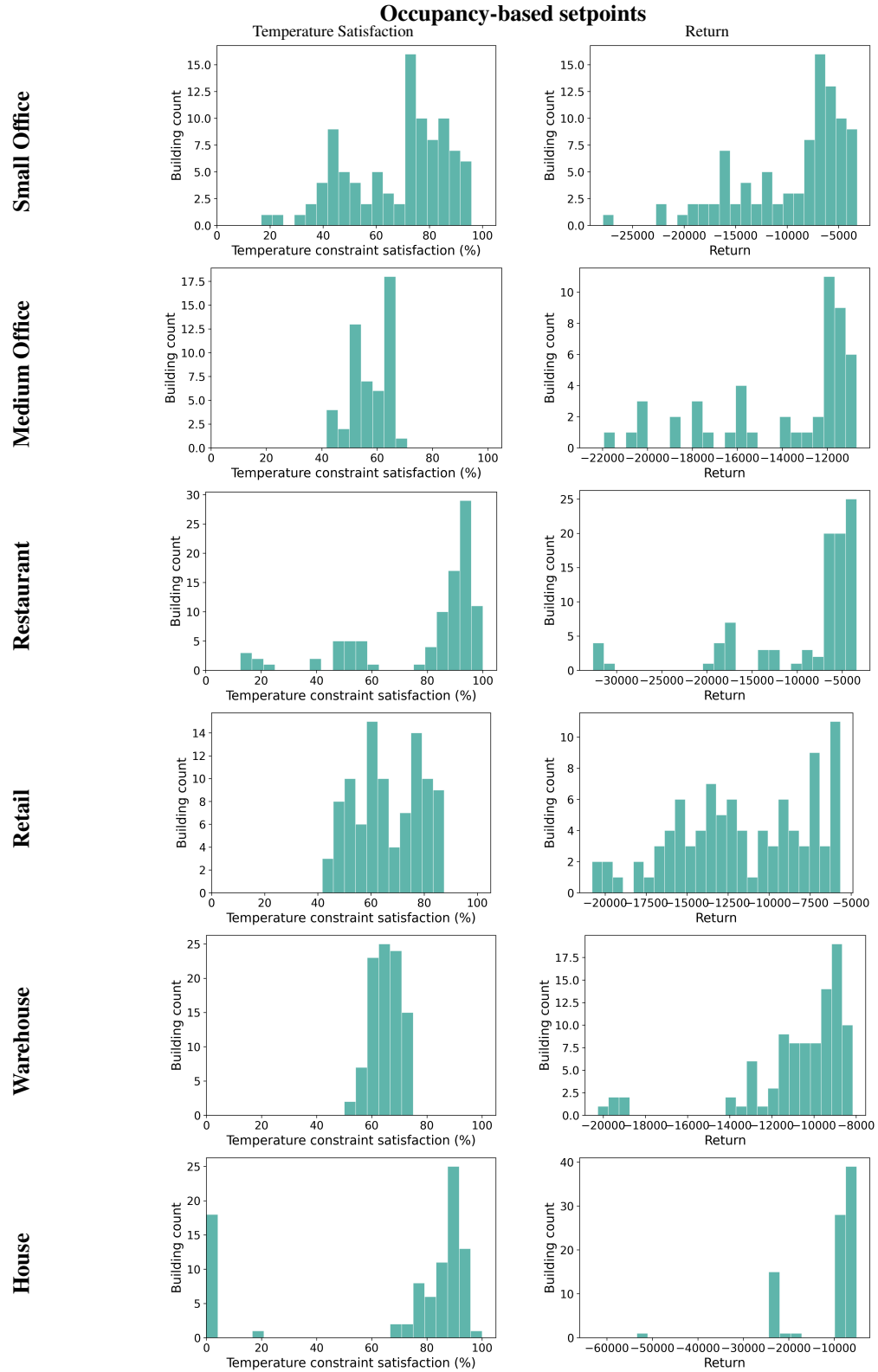
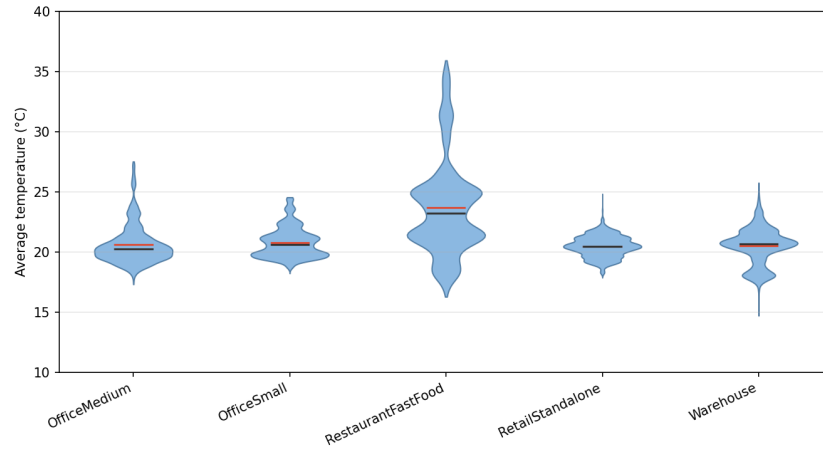
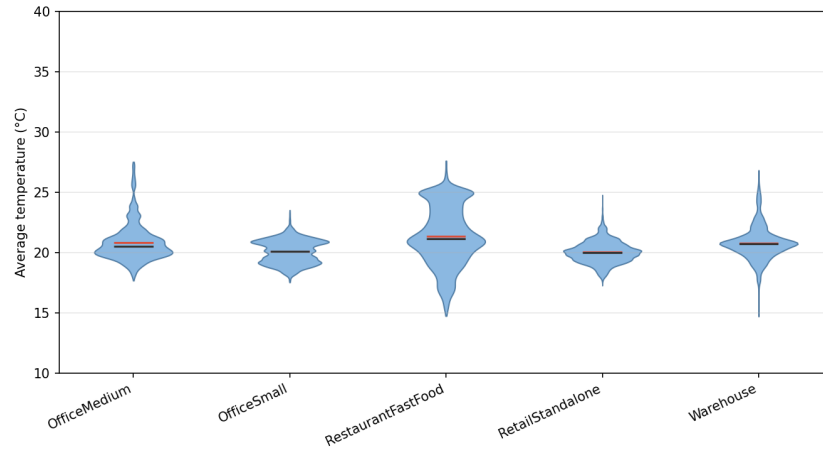


Figure 5: Histograms of temperature satisfaction and returns for occupancy-based setpoints (task 3).



(a) Task 1



(b) Task 4

Figure 6: Temperature performance across test buildings under two tasks

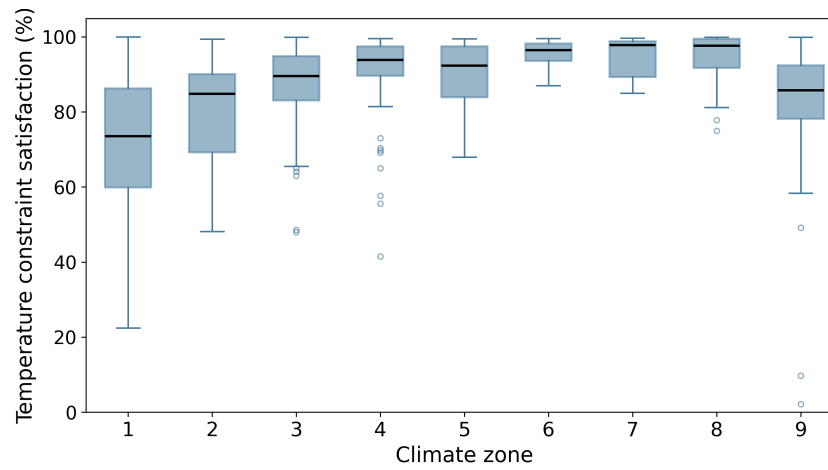


Figure 7: Influence of the climate zone on the control. Climate zones are ordered from hottest to coldest